Predictive QoS Aware PPO Routing for SDN

Computer Networks Project Report

MOHAMMED THALHA-25BDS0206

AVANI VASISTHA – 25BCE0941

PANGULURU ANEESH - 25BCE2328

Project area: Software Defined Networking and Reinforcement Learning

Implementation: Mininet, Ryu, Stable Baselines3 and Python

Evaluation: Simulation and live emulated network experiments

Github Link: https://github.com/aneeshpanguluru-oss/sdn-qos-ppo

Literature Review:

Ref.	Author(s), Year	Category / Theme	Proposed Technique	Dataset / Tools	Key Findings	Limitations
[1]	Wang et al. (2025)	Segment-routing traffic engineering	RS4SR: recommends promising routing candidates from historical solutions and clusters similar traffic flows to reduce the optimization search space	Large-scale network evaluations; specific topologies	Reported near-optimal solutions within a two-second time limit using a relatively small set of historical solutions	Accelerates mathematical optimization; performance under rapidly changing traffic conditions needs further study
[2]	Tian et al. (2025)	Adaptive segment-routing traffic engineering	Adpt-SRTE: PPO with action branching; learns from historical traffic matrices and uses current link utilization for online decisions	Three real-world network topologies	Reported up to 33% reduction in maximum link utilization and online decisions in tens of milliseconds	Focuses on segment routing; generalization to unfamiliar traffic patterns and other SDN settings requires study
[3]	Guo et al. (2025)	Persistent routing; supervised learning	PROM: predicts routing strategies from historical traffic matrices using candidate paths generated by a k-shortest-paths algorithm	Three real-world network topologies; centralized and distributed implementations	Reported effective routing strategies, with comparable performance from the distributed implementation	Depends on historical examples; adaptation to unexpected traffic changes needs investigation
[4]	Zhou et al. (2025)	Hybrid SDN traffic engineering; representation learning	CRTE: combines contrastive traffic representation learning with reinforcement learning; uses K-means to form training samples	Hybrid SDN evaluation; topology and tool details	Reported reductions in maximum link utilization and end-to-end latency, and robustness under network failures	Encoder, RL algorithm, topology, and baseline details should be verified in the full paper
[5]	Chen et al. (2025)	SD-WAN traffic engineering; multi-QoS routing	MTEAL: extends TEAL with multi-dimensional QoS measures and a two-stage reward function	Small, NSFNET, and GBN topologies; comparisons include UsCarrier, Kdl, and B4	Reported improvements over comparison methods, including results for delay, flow loss, failure stability, and generalization	Exact reward formula and topology-specific numerical results should be checked in the full article
[6]	Wu et al. (2026)	SDN routing; topology-aware DRL	IROD-BC: combines DRL routing with betweenness centrality to guide optimization	Specific topologies, tools, and implementation details	Intended to guide routing using network-structure information and address traffic concentration on important network elements	Available summary does not establish the precise algorithm, centrality calculation, baselines, or quantitative gains
[7]	Awad et al. (2026)	SDN routing; transfer learning	DDPG-DR: applies transfer learning to a DDPG routing agent so it can reuse prior experience after topology changes or link failures	Ryu controller and Mininet; Abilene topology; traffic loads from 20% to 80%	Reported higher throughput than OSPF; at 80% load, 39 Mbps versus 24.7 Mbps	Evaluation summary covers a limited topology and load range; exact delay comparisons should be checked against the paper
[8]	Lin et al. (2025)	SDN routing; bandwidth-demand-aware path selection	Hop-by-hop DRL routing considers flow bandwidth demand and available link capacity; uses prioritized experience replay	Ryu, Open vSwitch, Docker, and Mininet; tests include 9-switch and 34-node networks and larger topologies	Reported 39.95% bandwidth utilization versus 36.35% for Q-WPRA; inference time of about 1.05–2.85 ms across tested topologies	Reported experiments use particular traffic and network scenarios; broader evaluations are needed to establish generality

Ref.	Author(s), Year	Category / Theme	Proposed Technique	Dataset / Tools	Key Findings	Limitations
[9]	Aguirre S. et al. (2025)	SDN congestion avoidance; QoS-aware routing	LCA: classifies traffic, monitors live QoS and congestion, selects routes with DRL, and validates routes	Ryu controller and Docker-based networking; ten service classes; classifier trained on 500,000 flows	Reported 5–10% lower delay, 5–20% lower packet loss, and 25–30% lower jitter under congestion	Classifier, monitoring, and validation add system components; performance outside the evaluated setup is not established by the summary

Research Gaps

1. **Limited testing on unfamiliar traffic and topologies:**

Many learning-based methods use historical traffic or a limited set of network topologies. Their ability to generalize to unseen traffic patterns, new topologies, or sudden network changes remains insufficiently established.

2. **Need for faster adaptation to failures:**

Some papers investigate link failures or transfer learning, but more work is needed to evaluate how quickly routing agents recover from failures and whether they can adapt without extensive retraining.

3. **Incomplete multi-objective evaluation:**

Several studies focus primarily on congestion, link utilization, or throughput. A more complete evaluation should consider these alongside delay, packet loss, jitter, bandwidth demand, and routing overhead.

4. **Limited comparison of traffic representations:**

Methods use different input information, such as raw traffic measurements, learned traffic features, link utilization, or topology measures. The effect of these representations on routing performance has not been compared consistently.

5. **Unclear deployment and scalability trade-offs:**

Although some studies report quick routing decisions, further evaluation is needed on controller workload, communication overhead, scalability to larger networks, and performance in live or larger-scale deployments.

6. **Limited connection between predictive and SDN routing:**

Predictive congestion avoidance has been studied in AAV networks, while most SDN papers react to current network conditions. The usefulness of predicting future traffic and congestion for SDN routing remains an opportunity for investigation.

Abstract

This project develops and evaluates a quality of service aware routing controller for software defined networks. The controller selects one of several paths between hosts h1 and h4 using Proximal Policy Optimization (PPO). It observes link utilization, path delay, packet loss estimates, residual capacity, and flow demand. A predictive PPO variant adds a one interval utilization forecast. The study compares both PPO policies with fixed shortest delay and min hop baselines in a controlled simulator and in a live Mininet network controlled by Ryu.

The central result is that PPO policies maintain high admitted throughput in the simulator and successfully reroute live traffic after an s2 to s4 link failure. Fixed Min hop fails when its selected route contains the broken link. The shortest delay baseline avoids that failed link but is constrained by a 15 Mbps bottleneck on its selected route.

1. Project Objective

The objective is to determine whether a reinforcement learning based SDN controller can make better route selections than fixed routing rules when link capacity, delay, background load, and failures affect quality of service.

- Train a PPO agent to choose among three candidate paths between s1 and s4.
- Compare PPO with fixed shortest delay and min hop routing rules.
- Evaluate throughput, maximum utilization, delay, loss, feasibility, decision time, and route churn in simulation.
- Validate the controller in Mininet with Ryu and iperf3 under normal traffic and an injected link failure.

2. System Design

The network uses a six switch Mininet topology with hosts h1 and h4 attached to the experiment switches. The controller selects one of three predefined candidate routes. Explicit switch identifiers and port mappings let Ryu install OpenFlow flow rules and measure physical port utilization.

Candidate	Route	Main capacity constraint	Approximate delay
0	s1 - s2 - s4	25 Mbps on s2-s4	10 ms
1	s1 - s3 - s4	20 Mbps on s1-s3	11 ms
2	s1 - s2 - s3 - s4	15 Mbps on s2-s3	9 ms

Table 1. Candidate paths used by the routing action space.

3. Methods

3.1 PPO Routing State and Reward

For each candidate path, the standard PPO agent receives normalized maximum utilization, delay, loss estimate, and residual capacity. The final state feature is the incoming flow demand. The predictive agent adds a momentum based utilization forecast. The reward favors admitted throughput and residual capacity while penalizing high utilization, delay, loss, and unnecessary route changes.

3.2 Baselines

- Shortest delay: always selects candidate 2 because it has the lowest fixed delay.
- Min hop: always selects candidate 0 because it has the fewest switches.
- PPO standard QoS: policy trained on current QoS features.
- PPO predictive QoS: policy trained with the added utilization forecast feature.

3.3 Failure Model

The simulation disables the s2-s4 link halfway through each episode. In the live environment, the same fault is injected with the Mininet command link s2 s4 down. PPO modes receive the port down event, remove

stale managed flows, and install a viable route. Fixed baselines have automatic recovery disabled to keep the comparison fair.

4. Simulation Results

Both PPO policies were trained for 100,000 timesteps and evaluated across 50 simulated episodes of 120 steps each. The values below are mean results from the project evaluation program. Higher throughput and feasibility are better; lower maximum utilization, delay, loss, decision time, and route churn are better.

4.1 Normal Traffic Scenario

Policy	Throughput Mbps	Max Util	Delay ms	Loss	Feasible	Decision us	Churn per ep
Shortest path delay	1.76	0.665	9.00	1.61%	28.7%	4.3	0.0
Min hop	3.40	0.757	10.00	1.67%	42.6%	1.3	0.0
PPO standard QoS	5.70	0.667	10.27	1.29%	65.2%	252.0	69.4
PPO predictive QoS	5.62	0.659	10.38	1.24%	63.9%	227.4	70.4

Table 2. Simulation results under normal traffic.

Interpretation. Standard PPO achieves the highest simulated throughput and feasibility. Predictive PPO has the lowest average utilization and loss, but it does not outperform standard PPO on throughput in this training run. Both PPO policies substantially outperform the fixed baselines on throughput and feasibility.

4.2 Link Failure Scenario

Policy	Throughput Mbps	Max Util	Delay ms	Loss	Feasible	Decision us	Churn per ep
Shortest path delay	1.76	0.665	9.00	1.61%	28.7%	5.1	0.0
Min hop	1.74	0.881	10.00	50.96%	21.4%	1.0	0.0
PPO standard QoS	4.83	0.694	10.07	1.53%	57.0%	238.3	60.9
PPO predictive QoS	4.67	0.690	10.36	1.51%	55.8%	240.8	60.1

Table 3. Simulation results when the s2-s4 link fails at step 60.

Interpretation. Min hop is heavily affected because its fixed route uses s2-s4. Its simulated loss reaches 50.96 percent. Both PPO policies preserve much higher throughput and feasibility by selecting viable alternatives. The shortest delay route is unchanged because it does not use the failed s2-s4 link, but it remains limited by the lower capacity s2-s3 link.

5. Live Ryu and Mininet Validation

The Ryu controller was connected to the Mininet topology and monitored port statistics. Traffic was generated with iperf3 using a 20 Mbps UDP offered load. The fault experiment lowered the s2-s4 link after approximately ten seconds. PPO controller logs confirmed the selected candidate path and, on failure, reported recovery to candidate 1: s1 - s3 - s4.

5.1 Observed Live Normal Traffic Results

Policy	Selected route	Received throughput	Loss

Min hop	s1-s2-s4	19.9Mbps	0.38%
Shortest delay	s1-s2-s3-s4	14.1 Mbps	27%
PPO standard QoS	s1-s2-s4	19.9 Mbps	0.4%
PPO predictive QoS	s1-s2-s4	18,7 Mbps	0.44

Table 4. Live Ryu and Mininet normal traffic observations, 20 Mbps UDP offered load.

The measured normal-traffic results show why fixed shortest-delay routing is not automatically the best QoS choice: its selected route has a 15 Mbps bottleneck and therefore drops traffic at a 20 Mbps offered load. The standard PPO controller selected the higher-capacity route and delivered 19.9 Mbps with 0.36 percent loss.

5.2 Observed Live Link Failure Results

Policy	Path behavior after s2-s4 failure	Received throughput	Loss	Outcome
Min hop	Fixed path s1-s2-s4 becomes unavailable	5.02 Mbps*	About 61.1%*	Route failure
Shortest delay	Continues on s1-s2-s3-s4	14.5 Mbps	26%	15 Mbps bottleneck
PPO standard QoS	Reroutes to s1-s3-s4	19.5 Mbps	1.6%	Recovered
PPO predictive QoS	Reroutes to s1-s3-s4	19.5 Mbps	1.5%	Recovered

*Table 5. Live Ryu and Mininet fault test, 20 Mbps UDP offered load for 60 seconds. *For Min hop, iperf3 receiver reporting was disrupted by the route break. The loss estimate is derived from 103,590 sender datagrams and 40,344 receiver datagrams.*

The live test supports the simulation findings. Both PPO controllers continued delivering approximately 19.5 Mbps after the failed link and incurred only around one to two percent loss. Min hop lost its selected route. Shortest delay did not use the failed link but was limited by its 15 Mbps bottleneck, so it achieved 14.5 Mbps with 26 percent loss.

6. Commands Used

Simulation comparison command:

```
PYTHONPATH=src python -m sdn_rl.evaluate --compare-all --episodes 50
```

Live controller modes:

```
PYTHONPATH=src ROUTING_MODE=ppo ryu-manager sdn_rl.ryu_app
PYTHONPATH=src ROUTING_MODE=predictive ryu-manager sdn_rl.ryu_app
PYTHONPATH=src ROUTING_MODE=minhop ryu-manager sdn_rl.ryu_app
PYTHONPATH=src ROUTING_MODE=shortest ryu-manager sdn_rl.ryu_app
```

Fault injection in the Mininet CLI: start a persistent h4 iperf3 server, start a 60 second h1 UDP client, wait approximately ten seconds, then run link s2 s4 down. Restore with link s2 s4 up after the test.

7. Conclusion

The project demonstrates an end to end SDN traffic engineering workflow: a Gymnasium environment trains PPO policies, Stable Baselines3 produces the models, Ryu applies policy decisions as OpenFlow paths, and Mininet validates behavior under controlled traffic and faults. In simulation, PPO improves throughput and feasible decisions compared with fixed policies. In the live fault test, standard and predictive PPO

reroute successfully and maintain near offered load, while Min hop fails on the broken path and shortest delay is constrained by capacity. The next practical extension is multi flow routing with a broader set of source and destination pairs.

Proofs:

Training Predictive and Non Predictive PPO:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ export PYTHONPATH="$PWD/src:$PYTHONPATH"
python3 -m sdn_rl.train --timesteps 50000
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they up
grade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/.venv-ryu/lib/python3.11/site-packages/torch/cuda/__init__.py:228: UserWarning: CUDA
initialization: The NVIDIA driver on your system is too old (found version 12080). Please update your GPU driver by downloading and inst
alling a new version from the URL: http://www.nvidia.com/Download/index.aspx Alternatively, go to: https://pytorch.org to install a PyTo
rch version that has been compiled with your version of the CUDA driver. (Triggered internally at /__w/pytorch/pytorch/c10/cuda/CUDAFunc
tions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
Using cpu device
---------------------------------
| rollout/           |          |
|    ep_len_mean     | 120      |
|    ep_rew_mean     | -89.7    |
| time/              |          |
|    fps             | 2777     |
|    iterations      | 1        |
|    time_elapsed    | 0        |
|    total_timesteps | 2048     |
---------------------------------
-----------------------------------------
| rollout/                |             |
|    ep_len_mean          | 120         |
|    ep_rew_mean          | -89.8       |
| time/                   |             |
|    fps                  | 1943        |
|    iterations           | 2           |
|    time_elapsed         | 2           |
|    total_timesteps      | 4096        |
| train/                  |             |
|    approx_kl            | 0.010790731 |
|    clip_fraction        | 0.0734      |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.09       |
```

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ export PYTHONPATH="$PWD/src:$PYTHONPATH"
python3 -m sdn_rl.train --timesteps 50000 --predictive
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they up
grade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/.venv-ryu/lib/python3.11/site-packages/torch/cuda/__init__.py:228: UserWarning: CUDA
initialization: The NVIDIA driver on your system is too old (found version 12080). Please update your GPU driver by downloading and inst
alling a new version from the URL: http://www.nvidia.com/Download/index.aspx Alternatively, go to: https://pytorch.org to install a PyTo
rch version that has been compiled with your version of the CUDA driver. (Triggered internally at /__w/pytorch/pytorch/c10/cuda/CUDAFunc
tions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
Using cpu device
--------------------------------
| rollout/           |         |
|    ep_len_mean     | 120     |
|    ep_rew_mean     | -94.1   |
| time/              |         |
|    fps             | 2085    |
|    iterations      | 1       |
|    time_elapsed    | 0       |
|    total_timesteps | 2048    |
--------------------------------
-----------------------------------------
| rollout/                |             |
|    ep_len_mean          | 120         |
|    ep_rew_mean          | -88.3       |
| time/                   |             |
|    fps                  | 1635        |
|    iterations           | 2           |
|    time_elapsed         | 2           |
|    total_timesteps      | 4096        |
| train/                  |             |
|    approx_kl            | 0.010684533 |
|    clip_fraction        | 0.132       |
|    clip_range           | 0.2         |
|    entropy_loss         | -1.09       |
|    explained_variance   | 0.0113      |
```

Simulated Results:

```
(venv) PS C:\Users\QS\Documents\casestudy_cn\sdn-qos-ppo> python -m sdn_rl.evaluate --model models/ppo_qos_router.zip --compare-all

Normal traffic scenario
Policy                  Throughput   Max util      Delay       Loss   Feasible      Decision   Churn/ep
Shortest path (delay)        1.76M      0.665     9.00ms      1.61%      28.7%        6.4us        0.0
Min hop                      3.40M      0.757    10.00ms      1.67%      42.6%        1.5us        0.0
PPO standard QoS             5.75M      0.666    10.18ms      1.36%      65.4%      286.4us       73.6
PPO predictive QoS           5.69M      0.669    10.10ms      1.42%      64.5%      322.9us       77.6

Fault tolerance: s2-s4 fails at step 60
Policy                  Throughput   Max util      Delay       Loss   Feasible      Decision   Churn/ep
Shortest path (delay)        1.76M      0.665     9.00ms      1.61%      28.7%        6.3us        0.0
Min hop                      1.74M      0.881    10.00ms     50.96%      21.4%        1.6us        0.0
PPO standard QoS             4.70M      0.688    10.02ms      1.54%      55.8%      348.8us       61.4
PPO predictive QoS           4.64M      0.689     9.83ms      1.60%      54.5%      347.3us       63.9
(venv) PS C:\Users\QS\Documents\casestudy_cn\sdn-qos-ppo>
```

No Link Failure Results:

1) Shortest Delay:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ export PYTHONPATH="$PWD/src:$PYTHONPATH"
ROUTING_MODE=shortest FLOW_DEMAND_MBPS=20 python3 -c "import eventlet; eventlet.monkey_patch(); from ryu.cmd.manager import main; main()" --of
p-tcp-listen-port 6653 sdn_rl.ryu_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: shortest
Automatic recovery: disabled
Lowest-delay baseline selected; PPO model loading skipped
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 2: [1, 2, 3, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 2: [4, 3, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 2: [1, 2, 3, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 2: [4, 3, 2, 1]
```

```
PS C:\Users\QS\Documents\casestudy_cn> wsl
[  5] local 10.0.0.1 port 34410 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  1.88 MBytes  15.7 Mbits/sec    8   39.6 KBytes
[  5]   1.00-2.00   sec  1.38 MBytes  11.5 Mbits/sec    8   17.0 KBytes
[  5]   2.00-3.00   sec   768 KBytes  6.29 Mbits/sec    8   11.3 KBytes
[  5]   3.00-4.00   sec  1.00 MBytes  8.39 Mbits/sec    2   26.9 KBytes
[  5]   4.00-5.00   sec   896 KBytes  7.34 Mbits/sec    5   24.0 KBytes
[  5]   5.00-6.00   sec  1.12 MBytes  9.44 Mbits/sec    2   43.8 KBytes
[  5]   6.00-7.00   sec  1.38 MBytes  11.5 Mbits/sec    8   32.5 KBytes
[  5]   7.00-8.00   sec  1.38 MBytes  11.5 Mbits/sec    2   45.2 KBytes
[  5]   8.00-9.00   sec  1.00 MBytes  8.39 Mbits/sec    7   25.5 KBytes
[  5]   9.00-10.00  sec  1.25 MBytes  10.5 Mbits/sec    2   43.8 KBytes
[  5]  10.00-11.00  sec  1.12 MBytes  9.44 Mbits/sec    6   35.4 KBytes
[  5]  11.00-12.00  sec  1.75 MBytes  14.7 Mbits/sec    6   33.9 KBytes
[  5]  12.00-13.00  sec   768 KBytes  6.29 Mbits/sec   12   15.6 KBytes
[  5]  13.00-14.00  sec   768 KBytes  6.29 Mbits/sec    4   22.6 KBytes
[  5]  14.00-15.00  sec   896 KBytes  7.34 Mbits/sec    5   21.2 KBytes
[  5]  15.00-16.00  sec  1.00 MBytes  8.39 Mbits/sec    5   18.4 KBytes
[  5]  16.00-17.00  sec  1.12 MBytes  9.43 Mbits/sec    2   36.8 KBytes
[  5]  17.00-18.00  sec  1.12 MBytes  9.44 Mbits/sec    8   17.0 KBytes
[  5]  18.00-19.00  sec  1.00 MBytes  8.39 Mbits/sec    0   48.1 KBytes
[  5]  19.00-20.00  sec  1.38 MBytes  11.5 Mbits/sec    8   18.4 KBytes
[  5]  20.00-21.00  sec  1.00 MBytes  8.39 Mbits/sec    4   29.7 KBytes
[  5]  21.00-22.00  sec  1.25 MBytes  10.5 Mbits/sec    4   29.7 KBytes
[  5]  22.00-23.00  sec  1.38 MBytes  11.5 Mbits/sec    4   33.9 KBytes
[  5]  23.00-24.00  sec   768 KBytes  6.29 Mbits/sec   12   15.6 KBytes
[  5]  24.00-25.00  sec  1.12 MBytes  9.44 Mbits/sec    1   31.1 KBytes
[  5]  25.00-26.00  sec  1.12 MBytes  9.44 Mbits/sec    8   24.0 KBytes
[  5]  26.00-27.00  sec  1.25 MBytes  10.5 Mbits/sec    6   26.9 KBytes
[  5]  27.00-28.00  sec   896 KBytes  7.34 Mbits/sec    8   17.0 KBytes
[  5]  28.00-29.00  sec   768 KBytes  6.29 Mbits/sec    4   25.5 KBytes
[  5]  29.00-30.00  sec  1.12 MBytes  9.42 Mbits/sec    4   29.7 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  33.5 MBytes  9.37 Mbits/sec  163             sender
[  5]   0.00-30.03  sec  33.1 MBytes  9.25 Mbits/sec                  receiver

iperf Done.
```

2) Min Hop:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ ROUTING_MODE=minhop FLOW_DEMAND_MBPS=20 python3 -c "import
 eventlet; eventlet.monkey_patch(); from ryu.cmd.manager import main; main()" --ofp-tcp-listen-port 6653 sdn_rl.ryu_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: minhop
Automatic recovery: disabled
Min-hop baseline selected; PPO model loading skipped
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
```

```
PS C:\Users\QS\Documents\casestudy_cn> wsl
[  5] local 10.0.0.1 port 34588 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  2.50 MBytes  21.0 Mbits/sec    2    102 KBytes
[  5]   1.00-2.00   sec  2.38 MBytes  19.9 Mbits/sec    6   65.0 KBytes
[  5]   2.00-3.00   sec  2.38 MBytes  19.9 Mbits/sec    2   65.0 KBytes
[  5]   3.00-4.00   sec  2.38 MBytes  19.9 Mbits/sec    2   72.1 KBytes
[  5]   4.00-5.00   sec  2.38 MBytes  19.9 Mbits/sec    7   24.0 KBytes
[  5]   5.00-6.00   sec  1.00 MBytes  8.39 Mbits/sec    3   29.7 KBytes
[  5]   6.00-7.00   sec  1.25 MBytes  10.5 Mbits/sec    2   42.4 KBytes
[  5]   7.00-8.00   sec  1.50 MBytes  12.6 Mbits/sec    6   26.9 KBytes
[  5]   8.00-9.00   sec  1.25 MBytes  10.5 Mbits/sec    2   42.4 KBytes
[  5]   9.00-10.00  sec  1.75 MBytes  14.7 Mbits/sec    6   35.4 KBytes
[  5]  10.00-11.00  sec  1.00 MBytes  8.39 Mbits/sec    4   35.4 KBytes
[  5]  11.00-12.00  sec  1.50 MBytes  12.6 Mbits/sec    4   36.8 KBytes
[  5]  12.00-13.00  sec  1.50 MBytes  12.6 Mbits/sec    1   50.9 KBytes
[  5]  13.00-14.00  sec  1.75 MBytes  14.7 Mbits/sec    6   35.4 KBytes
[  5]  14.00-15.00  sec  1.50 MBytes  12.6 Mbits/sec    8   17.0 KBytes
[  5]  15.00-16.00  sec   768 KBytes  6.29 Mbits/sec    6   17.0 KBytes
[  5]  16.00-17.00  sec  1.00 MBytes  8.39 Mbits/sec    2   29.7 KBytes
[  5]  17.00-18.00  sec  1.62 MBytes  13.6 Mbits/sec    2   39.6 KBytes
[  5]  18.00-19.00  sec  2.00 MBytes  16.8 Mbits/sec    2   46.7 KBytes
[  5]  19.00-20.00  sec  2.25 MBytes  18.9 Mbits/sec    0   74.9 KBytes
[  5]  20.00-21.00  sec  2.75 MBytes  23.1 Mbits/sec    2   72.1 KBytes
[  5]  21.00-22.00  sec  2.25 MBytes  18.9 Mbits/sec    5   53.7 KBytes
[  5]  22.00-23.00  sec  2.00 MBytes  16.8 Mbits/sec    8   31.1 KBytes
[  5]  23.00-24.00  sec  1.25 MBytes  10.5 Mbits/sec    4   29.7 KBytes
[  5]  24.00-25.00  sec  1.75 MBytes  14.7 Mbits/sec    4   26.9 KBytes
[  5]  25.00-26.00  sec  1.50 MBytes  12.6 Mbits/sec    2   38.2 KBytes
[  5]  26.00-27.00  sec  1.50 MBytes  12.6 Mbits/sec    4   32.5 KBytes
[  5]  27.00-28.00  sec  1.00 MBytes  8.39 Mbits/sec    3   38.2 KBytes
[  5]  28.00-29.00  sec   768 KBytes  6.29 Mbits/sec   10   12.7 KBytes
[  5]  29.00-30.00  sec  1.00 MBytes  8.37 Mbits/sec    0   39.6 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  49.4 MBytes  13.8 Mbits/sec  115             sender
[  5]   0.00-30.03  sec  48.9 MBytes  13.7 Mbits/sec                  receiver

iperf Done.
```

3) Non Predictive PPO:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ ROUTING_MODE=ppo FLOW_DEMAND_MBPS=20 python3 -c "import ev
entlet; eventlet.monkey_patch(); from ryu.cmd.manager import main; main()" --ofp-tcp-listen-port 6653 sdn_rl.ryu_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: ppo
Automatic recovery: enabled
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they upgrade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/.venv-ryu/lib/python3.11/site-packages/torch/cuda/__init__.py:228: UserWarning: CUDA initia
lization: The NVIDIA driver on your system is too old (found version 12080). Please update your GPU driver by downloading and installing a new
 version from the URL: http://www.nvidia.com/Download/index.aspx Alternatively, go to: https://pytorch.org to install a PyTorch version that h
as been compiled with your version of the CUDA driver. (Triggered internally at /__w/pytorch/pytorch/c10/cuda/CUDAFunctions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
Loaded PPO policy from /mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/models/ppo_qos_router.zip
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
```

```
Connecting to host 10.0.0.4, port 5201
[  5] local 10.0.0.1 port 58914 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  2.50 MBytes  21.0 Mbits/sec    7   39.6 KBytes
[  5]   1.00-2.00   sec  1.50 MBytes  12.6 Mbits/sec    4   35.4 KBytes
[  5]   2.00-3.00   sec  1.50 MBytes  12.6 Mbits/sec    6   31.1 KBytes
[  5]   3.00-4.00   sec  1.00 MBytes  8.38 Mbits/sec    6   24.0 KBytes
[  5]   4.00-5.00   sec   768 KBytes  6.29 Mbits/sec    8   17.0 KBytes
[  5]   5.00-6.00   sec   768 KBytes  6.29 Mbits/sec    5   28.3 KBytes
[  5]   6.00-7.00   sec  1.00 MBytes  8.39 Mbits/sec    6   32.5 KBytes
[  5]   7.00-8.00   sec  1.25 MBytes  10.5 Mbits/sec    2   45.2 KBytes
[  5]   8.00-9.00   sec  1.50 MBytes  12.6 Mbits/sec    4   41.0 KBytes
[  5]   9.00-10.00  sec  1.50 MBytes  12.6 Mbits/sec   10   24.0 KBytes
[  5]  10.00-11.00  sec  1.00 MBytes  8.39 Mbits/sec    4   29.7 KBytes
[  5]  11.00-12.00  sec  1.25 MBytes  10.5 Mbits/sec    2   43.8 KBytes
[  5]  12.00-13.00  sec  1.50 MBytes  12.6 Mbits/sec    4   43.8 KBytes
[  5]  13.00-14.00  sec  1.50 MBytes  12.6 Mbits/sec    4   39.6 KBytes
[  5]  14.00-15.00  sec  1.25 MBytes  10.5 Mbits/sec    4   35.4 KBytes
[  5]  15.00-16.00  sec  1.12 MBytes  9.44 Mbits/sec    4   39.6 KBytes
[  5]  16.00-17.00  sec  1.25 MBytes  10.5 Mbits/sec    6   25.5 KBytes
[  5]  17.00-18.00  sec  1.50 MBytes  12.6 Mbits/sec    0   53.7 KBytes
[  5]  18.00-19.00  sec  1.50 MBytes  12.6 Mbits/sec    6   33.9 KBytes
[  5]  19.00-20.00  sec  1.50 MBytes  12.6 Mbits/sec    2   42.4 KBytes
[  5]  20.00-21.00  sec  1.50 MBytes  12.6 Mbits/sec    4   41.0 KBytes
[  5]  21.00-22.00  sec  1.00 MBytes  8.39 Mbits/sec    5   29.7 KBytes
[  5]  22.00-23.00  sec  1.50 MBytes  12.6 Mbits/sec    3   42.4 KBytes
[  5]  23.00-24.00  sec  1.50 MBytes  12.6 Mbits/sec    2   55.1 KBytes
[  5]  24.00-25.00  sec  2.75 MBytes  23.1 Mbits/sec    2   59.4 KBytes
[  5]  25.00-26.00  sec  1.75 MBytes  14.7 Mbits/sec    6   33.9 KBytes
[  5]  26.00-27.00  sec  1.00 MBytes  8.39 Mbits/sec    4   36.8 KBytes
[  5]  27.00-28.00  sec  1.50 MBytes  12.6 Mbits/sec    4   32.5 KBytes
[  5]  28.00-29.00  sec  1.50 MBytes  12.6 Mbits/sec    4   45.2 KBytes
[  5]  29.00-30.00  sec  1.62 MBytes  13.6 Mbits/sec    4   36.8 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  42.2 MBytes  11.8 Mbits/sec  132             sender
[  5]   0.00-30.03  sec  41.9 MBytes  11.7 Mbits/sec                  receiver

iperf Done.
```

4)Predictive PPO:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ ROUTING_MODE=predictive FLOW_DEMAND_MBPS=20 python3 -c "im
port eventlet; eventlet.monkey_patch(); from ryu.cmd.manager import main; main()" --ofp-tcp-listen-port 6653 sdn_rl.ryu_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: predictive
Automatic recovery: enabled
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they upgrade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/.venv-ryu/lib/python3.11/site-packages/torch/cuda/__init__.py:228: UserWarning: CUDA initia
lization: The NVIDIA driver on your system is too old (found version 12080). Please update your GPU driver by downloading and installing a new
 version from the URL: http://www.nvidia.com/Download/index.aspx Alternatively, go to: https://pytorch.org to install a PyTorch version that h
as been compiled with your version of the CUDA driver. (Triggered internally at /__w/pytorch/pytorch/c10/cuda/CUDAFunctions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
Loaded PPO policy from /mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/models/ppo_predictive_router.zip
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.4 -> 10.0.0.1 selected candidate 1: [4, 3, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 1: [1, 3, 4]
```

```
Connecting to host 10.0.0.4, port 5201
[  5] local 10.0.0.1 port 48584 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  2.50 MBytes  21.0 Mbits/sec    6   50.9 KBytes
[  5]   1.00-2.00   sec  1.25 MBytes  10.5 Mbits/sec    9   17.0 KBytes
[  5]   2.00-3.00   sec  1.00 MBytes  8.39 Mbits/sec    4   21.2 KBytes
[  5]   3.00-4.00   sec   768 KBytes  6.29 Mbits/sec    4   31.1 KBytes
[  5]   4.00-5.00   sec  1.75 MBytes  14.7 Mbits/sec    0   59.4 KBytes
[  5]   5.00-6.00   sec  2.25 MBytes  18.9 Mbits/sec    2   66.5 KBytes
[  5]   6.00-7.00   sec  1.75 MBytes  14.7 Mbits/sec    6   39.6 KBytes
[  5]   7.00-8.00   sec  1.00 MBytes  8.39 Mbits/sec    6   25.5 KBytes
[  5]   8.00-9.00   sec  1.50 MBytes  12.6 Mbits/sec    2   43.8 KBytes
[  5]   9.00-10.00  sec  1.75 MBytes  14.7 Mbits/sec    2   48.1 KBytes
[  5]  10.00-11.00  sec  1.50 MBytes  12.6 Mbits/sec    4   41.0 KBytes
[  5]  11.00-12.00  sec  1.50 MBytes  12.6 Mbits/sec    7   24.0 KBytes
[  5]  12.00-13.00  sec  1.00 MBytes  8.39 Mbits/sec    2   38.2 KBytes
[  5]  13.00-14.00  sec  1.75 MBytes  14.7 Mbits/sec    2   46.7 KBytes
[  5]  14.00-15.00  sec  1.25 MBytes  10.5 Mbits/sec    6   29.7 KBytes
[  5]  15.00-16.00  sec  1.25 MBytes  10.5 Mbits/sec    6   31.1 KBytes
[  5]  16.00-17.00  sec  1.12 MBytes  9.44 Mbits/sec    4   32.5 KBytes
[  5]  17.00-18.00  sec   768 KBytes  6.29 Mbits/sec    8   19.8 KBytes
[  5]  18.00-19.00  sec  1.00 MBytes  8.39 Mbits/sec    2   39.6 KBytes
[  5]  19.00-20.00  sec  1.75 MBytes  14.7 Mbits/sec    3   33.9 KBytes
[  5]  20.00-21.00  sec  1.25 MBytes  10.5 Mbits/sec    4   32.5 KBytes
[  5]  21.00-22.00  sec  1.00 MBytes  8.39 Mbits/sec    4   36.8 KBytes
[  5]  22.00-23.00  sec  1.00 MBytes  8.39 Mbits/sec    6   29.7 KBytes
[  5]  23.00-24.00  sec  1.50 MBytes  12.6 Mbits/sec    2   41.0 KBytes
[  5]  24.00-25.00  sec  1.50 MBytes  12.6 Mbits/sec    5   25.5 KBytes
[  5]  25.00-26.00  sec  1.50 MBytes  12.6 Mbits/sec    1   36.8 KBytes
[  5]  26.00-27.00  sec  1.50 MBytes  12.6 Mbits/sec    2   48.1 KBytes
[  5]  27.00-28.00  sec  1.25 MBytes  10.5 Mbits/sec    5   33.9 KBytes
[  5]  28.00-29.00  sec   768 KBytes  6.29 Mbits/sec    5   26.9 KBytes
[  5]  29.00-30.00  sec  1.25 MBytes  10.5 Mbits/sec    1   41.0 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  40.9 MBytes  11.4 Mbits/sec  120             sender
[  5]   0.00-30.03  sec  40.4 MBytes  11.3 Mbits/sec                  receiver

iperf Done.
```

s2-s4 link failure Results:

1) Shortest Delay:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ export PYTHONPATH="$PWD/src:$PYTHONPATH"
ROUTING_MODE=shortest RECOVERY_ENABLED=1 FLOW_DEMAND_MBPS=20 python3 -c "import eventlet; eventlet.monkey_patch(); from ryu.cmd.manager
import main; main()" --ofp-tcp-listen-port 6653 sdn_rl.ryu_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: shortest
Automatic recovery: enabled
Lowest-delay baseline selected; PPO model loading skipped
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 2: [1, 2, 3, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 2: [4, 3, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 2: [1, 2, 3, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 2: [4, 3, 2, 1]
```

```
Connecting to host 10.0.0.4, port 5201
[  5] local 10.0.0.1 port 37820 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  2.00 MBytes  16.8 Mbits/sec    6   53.7 KBytes
[  5]   1.00-2.00   sec  1.38 MBytes  11.5 Mbits/sec    6   41.0 KBytes
[  5]   2.00-3.00   sec  1.50 MBytes  12.6 Mbits/sec    2   50.9 KBytes
[  5]   3.00-4.00   sec  1.25 MBytes  10.5 Mbits/sec   10   17.0 KBytes
[  5]   4.00-5.00   sec  1.00 MBytes  8.39 Mbits/sec    4   21.2 KBytes
[  5]   5.00-6.00   sec  1.50 MBytes  12.6 Mbits/sec    0   50.9 KBytes
[  5]   6.00-7.00   sec  1.25 MBytes  10.5 Mbits/sec    9   25.5 KBytes
[  5]   7.00-8.00   sec   768 KBytes  6.29 Mbits/sec    8   15.6 KBytes
[  5]   8.00-9.00   sec   768 KBytes  6.29 Mbits/sec    4   25.5 KBytes
[  5]   9.00-10.00  sec  1.50 MBytes  12.6 Mbits/sec    2   41.0 KBytes
[  5]  10.00-11.00  sec  1.50 MBytes  12.6 Mbits/sec    2   55.1 KBytes
[  5]  11.00-12.00  sec  1.50 MBytes  12.6 Mbits/sec   11   18.4 KBytes
[  5]  12.00-13.00  sec  1.00 MBytes  8.39 Mbits/sec    2   31.1 KBytes
[  5]  13.00-14.00  sec  1.00 MBytes  8.39 Mbits/sec    8   18.4 KBytes
[  5]  14.00-15.00  sec   512 KBytes  4.20 Mbits/sec    8   24.0 KBytes
[  5]  15.00-16.00  sec  1.00 MBytes  8.39 Mbits/sec    4   32.5 KBytes
[  5]  16.00-17.00  sec  1.12 MBytes  9.44 Mbits/sec    6   24.0 KBytes
[  5]  17.00-18.00  sec  1.25 MBytes  10.5 Mbits/sec    4   31.1 KBytes
[  5]  18.00-19.00  sec  1.25 MBytes  10.5 Mbits/sec    6   21.2 KBytes
[  5]  19.00-20.00  sec  1.25 MBytes  10.5 Mbits/sec    6   17.0 KBytes
[  5]  20.00-21.00  sec  1.00 MBytes  8.39 Mbits/sec    4   24.0 KBytes
[  5]  21.00-22.00  sec  1.00 MBytes  8.39 Mbits/sec    6   33.9 KBytes
[  5]  22.00-23.00  sec  1.25 MBytes  10.5 Mbits/sec    6   24.0 KBytes
[  5]  23.00-24.00  sec  1.50 MBytes  12.6 Mbits/sec    0   52.3 KBytes
[  5]  24.00-25.00  sec  1.50 MBytes  12.6 Mbits/sec    6   31.1 KBytes
[  5]  25.00-26.00  sec  1.25 MBytes  10.5 Mbits/sec    4   33.9 KBytes
[  5]  26.00-27.00  sec   768 KBytes  6.29 Mbits/sec    8   35.4 KBytes
[  5]  27.00-28.00  sec  1.00 MBytes  8.39 Mbits/sec   10   12.7 KBytes
[  5]  28.00-29.00  sec   768 KBytes  6.29 Mbits/sec    4   24.0 KBytes
[  5]  29.00-30.00  sec  1.25 MBytes  10.4 Mbits/sec    4   26.9 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  35.5 MBytes  9.93 Mbits/sec  160             sender
[  5]   0.00-30.03  sec  35.0 MBytes  9.78 Mbits/sec                  receiver

iperf Done.
```

2) Min Hop:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ ROUTING_MODE=minhop RECOVERY_ENABLED=1 FLOW_DEMAND_M
BPS=20 python3 -c "import eventlet; eventlet.monkey_patch(); from ryu.cmd.manager import main; main()" --ofp-tcp-listen-port 6653 sdn_rl
.ryu_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: minhop
Automatic recovery: enabled
Min-hop baseline selected; PPO model loading skipped
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
```

```
Connecting to host 10.0.0.4, port 5201
[  5] local 10.0.0.1 port 41200 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  2.50 MBytes  21.0 Mbits/sec    5   52.3 KBytes
[  5]   1.00-2.00   sec  1.50 MBytes  12.6 Mbits/sec    4   45.2 KBytes
[  5]   2.00-3.00   sec  1.75 MBytes  14.7 Mbits/sec    2   58.0 KBytes
[  5]   3.00-4.00   sec  2.50 MBytes  21.0 Mbits/sec    2   60.8 KBytes
[  5]   4.00-5.00   sec  2.00 MBytes  16.8 Mbits/sec    6   31.1 KBytes
[  5]   5.00-6.00   sec  1.75 MBytes  14.7 Mbits/sec    0   59.4 KBytes
[  5]   6.00-7.00   sec  1.50 MBytes  12.6 Mbits/sec    4   53.7 KBytes
[  5]   7.00-8.00   sec  2.25 MBytes  18.9 Mbits/sec    6   36.8 KBytes
[  5]   8.00-9.00   sec  1.25 MBytes  10.5 Mbits/sec    4   25.5 KBytes
[  5]   9.00-10.00  sec  1.50 MBytes  12.6 Mbits/sec    2   38.2 KBytes
[  5]  10.00-11.00  sec  1.50 MBytes  12.6 Mbits/sec    2   50.9 KBytes
[  5]  11.00-12.00  sec  1.88 MBytes  15.7 Mbits/sec    2   59.4 KBytes
[  5]  12.00-13.00  sec  2.00 MBytes  16.8 Mbits/sec    3   45.2 KBytes
[  5]  13.00-14.00  sec  1.75 MBytes  14.7 Mbits/sec    4   41.0 KBytes
[  5]  14.00-15.00  sec  2.00 MBytes  16.8 Mbits/sec    2   49.5 KBytes
[  5]  15.00-16.00  sec  2.25 MBytes  18.9 Mbits/sec    2   52.3 KBytes
[  5]  16.00-17.00  sec  2.00 MBytes  16.8 Mbits/sec    6   42.4 KBytes
[  5]  17.00-18.00  sec  1.75 MBytes  14.7 Mbits/sec    4   33.9 KBytes
[  5]  18.00-19.00  sec  1.25 MBytes  10.5 Mbits/sec    3   33.9 KBytes
[  5]  19.00-20.00  sec  1.25 MBytes  10.5 Mbits/sec    4   33.9 KBytes
[  5]  20.00-21.00  sec  1.25 MBytes  10.5 Mbits/sec    1   48.1 KBytes
[  5]  21.00-22.00  sec  1.75 MBytes  14.7 Mbits/sec    7   26.9 KBytes
[  5]  22.00-23.00  sec  1.25 MBytes  10.5 Mbits/sec    2   39.6 KBytes
[  5]  23.00-24.00  sec  1.75 MBytes  14.7 Mbits/sec    4   33.9 KBytes
[  5]  24.00-25.00  sec  1.62 MBytes  13.6 Mbits/sec    2   45.2 KBytes
[  5]  25.00-26.00  sec  1.25 MBytes  10.5 Mbits/sec    4   42.4 KBytes
[  5]  26.00-27.00  sec  1.25 MBytes  10.5 Mbits/sec    4   45.2 KBytes
[  5]  27.00-28.00  sec  1.25 MBytes  10.5 Mbits/sec    4   43.8 KBytes
[  5]  28.00-29.00  sec  2.38 MBytes  19.9 Mbits/sec    1   52.3 KBytes
[  5]  29.00-30.00  sec  1.75 MBytes  14.7 Mbits/sec    4   42.4 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  51.6 MBytes  14.4 Mbits/sec  100             sender
[  5]   0.00-30.03  sec  51.1 MBytes  14.3 Mbits/sec                  receiver

iperf Done.
```

3) Non-predictive PPO:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ ROUTING_MODE=ppo RECOVERY_ENABLED=1 FLOW_DEMAND_MBPS
=20 python3 -c "import eventlet; eventlet.monkey_patch(); from ryu.cmd.manager import main; main()" --ofp-tcp-listen-port 6653 sdn_rl.ry
u_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: ppo
Automatic recovery: enabled
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they up
grade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/.venv-ryu/lib/python3.11/site-packages/torch/cuda/__init__.py:228: UserWarning: CUDA
initialization: The NVIDIA driver on your system is too old (found version 12080). Please update your GPU driver by downloading and inst
alling a new version from the URL: http://www.nvidia.com/Download/index.aspx Alternatively, go to: https://pytorch.org to install a PyTo
rch version that has been compiled with your version of the CUDA driver. (Triggered internally at /__w/pytorch/pytorch/c10/cuda/CUDAFunc
tions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
Loaded PPO policy from /mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/models/ppo_qos_router.zip
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.4 -> 10.0.0.1 selected candidate 1: [4, 3, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 1: [1, 3, 4]
```

```
Connecting to host 10.0.0.4, port 5201
[  5] local 10.0.0.1 port 44060 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  2.50 MBytes  21.0 Mbits/sec    5    129 KBytes
[  5]   1.00-2.00   sec  2.38 MBytes  19.9 Mbits/sec    5   73.5 KBytes
[  5]   2.00-3.00   sec  2.25 MBytes  18.9 Mbits/sec    8   26.9 KBytes
[  5]   3.00-4.00   sec  1.50 MBytes  12.6 Mbits/sec    0   56.6 KBytes
[  5]   4.00-5.00   sec  1.50 MBytes  12.6 Mbits/sec    7   24.0 KBytes
[  5]   5.00-6.00   sec   512 KBytes  4.19 Mbits/sec    7   19.8 KBytes
[  5]   6.00-7.00   sec  1.00 MBytes  8.39 Mbits/sec    4   24.0 KBytes
[  5]   7.00-8.00   sec   512 KBytes  4.19 Mbits/sec    7   15.6 KBytes
[  5]   8.00-9.00   sec  1.00 MBytes  8.39 Mbits/sec    4   21.2 KBytes
[  5]   9.00-10.00  sec  1.12 MBytes  9.44 Mbits/sec    2   35.4 KBytes
[  5]  10.00-11.00  sec  1.00 MBytes  8.39 Mbits/sec    4   35.4 KBytes
[  5]  11.00-12.00  sec  2.00 MBytes  16.8 Mbits/sec    4   43.8 KBytes
[  5]  12.00-13.00  sec  1.00 MBytes  8.39 Mbits/sec   10   21.2 KBytes
[  5]  13.00-14.00  sec  1.00 MBytes  8.39 Mbits/sec    2   36.8 KBytes
[  5]  14.00-15.00  sec  2.00 MBytes  16.8 Mbits/sec    0   66.5 KBytes
[  5]  15.00-16.00  sec  2.75 MBytes  23.1 Mbits/sec    2   67.9 KBytes
[  5]  16.00-17.00  sec  2.25 MBytes  18.9 Mbits/sec    8   38.2 KBytes
[  5]  17.00-18.00  sec  1.25 MBytes  10.5 Mbits/sec    2   48.1 KBytes
[  5]  18.00-19.00  sec  2.25 MBytes  18.9 Mbits/sec    2   53.7 KBytes
[  5]  19.00-20.00  sec  2.25 MBytes  18.9 Mbits/sec    2   60.8 KBytes
[  5]  20.00-21.00  sec  1.25 MBytes  10.5 Mbits/sec    6   38.2 KBytes
[  5]  21.00-22.00  sec  1.50 MBytes  12.6 Mbits/sec    4   36.8 KBytes
[  5]  22.00-23.00  sec  1.38 MBytes  11.5 Mbits/sec    4   33.9 KBytes
[  5]  23.00-24.00  sec  1.00 MBytes  8.39 Mbits/sec    5   25.5 KBytes
[  5]  24.00-25.00  sec  1.00 MBytes  8.39 Mbits/sec    4   29.7 KBytes
[  5]  25.00-26.00  sec  1.25 MBytes  10.5 Mbits/sec    2   43.8 KBytes
[  5]  26.00-27.00  sec  1.75 MBytes  14.7 Mbits/sec    4   38.2 KBytes
[  5]  27.00-28.00  sec  1.25 MBytes  10.5 Mbits/sec    4   33.9 KBytes
[  5]  28.00-29.00  sec  1.25 MBytes  10.5 Mbits/sec    4   32.5 KBytes
[  5]  29.00-30.00  sec  1.25 MBytes  10.5 Mbits/sec    5   21.2 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  44.9 MBytes  12.5 Mbits/sec  127             sender
[  5]   0.00-30.03  sec  44.4 MBytes  12.4 Mbits/sec                  receiver

iperf Done.
```

4)Predictive PPO:

```
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$
(.venv-ryu) thalha@HPProBook-g6:/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo$ ROUTING_MODE=predictive RECOVERY_ENABLED=1 FLOW_DEMA
ND_MBPS=20 python3 -c "import eventlet; eventlet.monkey_patch(); from ryu.cmd.manager import main; main()" --ofp-tcp-listen-port 6653 sd
n_rl.ryu_app
loading app sdn_rl.ryu_app
loading app ryu.controller.ofp_handler
instantiating app sdn_rl.ryu_app of QosPpoController
Routing mode: predictive
Automatic recovery: enabled
Gym has been unmaintained since 2022 and does not support NumPy 2.0 amongst other critical functionality.
Please upgrade to Gymnasium, the maintained drop-in replacement of Gym, or contact the authors of your software and request that they up
grade.
Users of this version of Gym should be able to simply replace 'import gym' with 'import gymnasium as gym' in the vast majority of cases.
See the migration guide at https://gymnasium.farama.org/introduction/migration_guide/ for additional information.
/mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/.venv-ryu/lib/python3.11/site-packages/torch/cuda/__init__.py:228: UserWarning: CUDA
initialization: The NVIDIA driver on your system is too old (found version 12080). Please update your GPU driver by downloading and inst
alling a new version from the URL: http://www.nvidia.com/Download/index.aspx Alternatively, go to: https://pytorch.org to install a PyTo
rch version that has been compiled with your version of the CUDA driver. (Triggered internally at /__w/pytorch/pytorch/c10/cuda/CUDAFunc
tions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
Loaded PPO policy from /mnt/c/Users/QS/Documents/casestudy_cn/sdn-qos-ppo/models/ppo_predictive_router.zip
instantiating app ryu.controller.ofp_handler of OFPHandler
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
10.0.0.1 -> 10.0.0.4 selected candidate 0: [1, 2, 4]
10.0.0.4 -> 10.0.0.1 selected candidate 0: [4, 2, 1]
```

```
Connecting to host 10.0.0.4, port 5201
[  5] local 10.0.0.1 port 57494 connected to 10.0.0.4 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  1.38 MBytes  11.5 Mbits/sec    4   35.4 KBytes
[  5]   1.00-2.00   sec  1.25 MBytes  10.5 Mbits/sec   10   15.6 KBytes
[  5]   2.00-3.00   sec  1.12 MBytes  9.44 Mbits/sec    0   42.4 KBytes
[  5]   3.00-4.00   sec  2.25 MBytes  18.9 Mbits/sec    0   72.1 KBytes
[  5]   4.00-5.00   sec  1.88 MBytes  15.7 Mbits/sec    8   25.5 KBytes
[  5]   5.00-6.00   sec  1.12 MBytes  9.44 Mbits/sec    2   42.4 KBytes
[  5]   6.00-7.00   sec  1.88 MBytes  15.7 Mbits/sec    2   52.3 KBytes
[  5]   7.00-8.00   sec  2.00 MBytes  16.8 Mbits/sec    4   41.0 KBytes
[  5]   8.00-9.00   sec  1.50 MBytes  12.6 Mbits/sec    5   36.8 KBytes
[  5]   9.00-10.00  sec  1.38 MBytes  11.5 Mbits/sec    8   15.6 KBytes
[  5]  10.00-11.00  sec  1.12 MBytes  9.44 Mbits/sec    2   29.7 KBytes
[  5]  11.00-12.00  sec  1.12 MBytes  9.44 Mbits/sec    2   43.8 KBytes
[  5]  12.00-13.00  sec  2.00 MBytes  16.8 Mbits/sec    6   36.8 KBytes
[  5]  13.00-14.00  sec  1.38 MBytes  11.5 Mbits/sec    6   33.9 KBytes
[  5]  14.00-15.00  sec  1.25 MBytes  10.5 Mbits/sec    4   32.5 KBytes
[  5]  15.00-16.00  sec  1.25 MBytes  10.5 Mbits/sec    2   43.8 KBytes
[  5]  16.00-17.00  sec  1.50 MBytes  12.6 Mbits/sec    6   26.9 KBytes
[  5]  17.00-18.00  sec  1.12 MBytes  9.44 Mbits/sec    4   29.7 KBytes
[  5]  18.00-19.00  sec  1.00 MBytes  8.39 Mbits/sec    4   36.8 KBytes
[  5]  19.00-20.00  sec  1.25 MBytes  10.5 Mbits/sec    4   35.4 KBytes
[  5]  20.00-21.00  sec  1.88 MBytes  15.7 Mbits/sec    0   65.0 KBytes
[  5]  21.00-22.00  sec  1.88 MBytes  15.7 Mbits/sec    6   35.4 KBytes
[  5]  22.00-23.00  sec  1.12 MBytes  9.44 Mbits/sec    4   33.9 KBytes
[  5]  23.00-24.00  sec  1.12 MBytes  9.43 Mbits/sec    8   25.5 KBytes
[  5]  24.00-25.00  sec  1.25 MBytes  10.5 Mbits/sec    2   36.8 KBytes
[  5]  25.00-26.00  sec  1.38 MBytes  11.5 Mbits/sec    4   35.4 KBytes
[  5]  26.00-27.00  sec  1.12 MBytes  9.44 Mbits/sec    6   25.5 KBytes
[  5]  27.00-28.00  sec  1.12 MBytes  9.44 Mbits/sec    4   29.7 KBytes
[  5]  28.00-29.00  sec   896 KBytes  7.34 Mbits/sec    4   33.9 KBytes
[  5]  29.00-30.00  sec  1.38 MBytes  11.5 Mbits/sec    4   49.5 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-30.00  sec  41.9 MBytes  11.7 Mbits/sec  125             sender
[  5]   0.00-30.03  sec  41.5 MBytes  11.6 Mbits/sec                  receiver

iperf Done.
```

References:

[1] L. Wang, M. Wang, C. Lin, and Y. Zhang, “Accelerating traffic engineering optimization for segment routing: A recommendation perspective,” *Computer Networks*, vol. 264, Art. no. 111224, 2025. https://doi.org/10.1016/j.comnet.2025.111224

[2] Y. Tian, Z. Wang, X. Yin, X. Shi, J. Yang, and H. Zhang, “Adaptive traffic engineering with segment routing through deep reinforcement learning,” *Computer Networks*, vol. 267, Art. no. 111356, 2025. https://doi.org/10.1016/j.comnet.2025.111356

[3] Y. Guo, Z. Huang, M. Ding, B. Lin, and H. Luo, “PROM: A persistent routing optimization method based on supervised learning,” *Journal of Network and Computer Applications*, vol. 242, Art. no. 104223, 2025. https://doi.org/10.1016/j.jnca.2025.104223

[4] W. Zhou, R. Yang, Y. Guo, and H. Luo, “Network traffic feature representation with contrastive learning for traffic engineering in hybrid software-defined networks,” *Journal of Network and Computer Applications*, vol. 242, Art. no. 104270, 2025. https://doi.org/10.1016/j.jnca.2025.104270

[5] Y. Chen, A. Gu, L. Cui, and L. Lin, “MTEAL: Network routing optimization of SD-WAN traffic engineering integrating multi-dimensional QoS metrics,” *Journal of Network and Computer Applications*, vol. 242, Art. no. 104272, 2025. https://doi.org/10.1016/j.jnca.2025.104272

[6] Z. Wu, Q. Tang, J. Cao, S. Wen, and B. Li, “Intelligent routing optimization with deep reinforcement learning and Betweenness Centrality Theory in software-defined networks,” *Computer Communications*, vol. 247, Art. no. 108401, 2026. https://doi.org/10.1016/j.comcom.2025.108401

[7] M. K. Awad, D. H. A. Daqar, D. H. Kablaoui, and E. A. Alrashed, “Knowledge transfer-enabled reinforcement learning for routing in software-defined networks,” *Journal of Engineering Research*, vol. 14, pp. 2840–2850, 2026. https://doi.org/10.1016/j.jer.2026.03.017

[8] G.-J. Lin, C.-F. Hung, and C.-H. Ke, “A deep reinforcement learning-based bandwidth demand-oriented routing in software-defined networking,” *ICT Express*, vol. 11, no. 6, pp. 1146–1151, 2025. https://doi.org/10.1016/j.icte.2025.07.009

[9] L. P. Aguirre S., Y. Shen, and M. Guo, “LCA: Deep reinforcement learning-based congestion avoidance routing model in SDN,” *Computer Networks*, vol. 268, Art. no. 111371, 2025. https://doi.org/10.1016/j.comnet.2025.111371

[10] R. I. Alkanhel, A. Alzaghir, N. D. Tu, A. Muthanna, A. Khakimov, and A. Koucheryavy, “Toward predictive routing in AAV network: A deep learning-based congestion avoidance approach,” *IEEE Access*, vol. 14, pp. 112202–112218, 2026. https://doi.org/10.1109/ACCESS.2026.3710875